

Business Report

UNSUPERVISED
LEARNING

Submitted By: Monika Baisla

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1. PROBLEM STATEMENT - PRODUCT SEGMENTATION

CONTEXT

When you think of sneakers for a trip, the importance of good footwear cannot be discarded, and the obvious brands that come to mind are Adidas and Nike. Adidas vs Nike is a constant debate as the two giants in the apparel market, with a large market cap and market share, battle it out to come on top. As a newly hired Data Scientist in a market research company, you have been given the task of extracting insights from the data of men's and women's shoes, and grouping products together to identify similarities and differences between the product range of these renowned brands.

OBJECTIVE

To perform an exploratory data analysis and cluster the products based on various factors.

KEY QUESTIONS

1. Which variables are most important for clustering?
2. How each cluster is different from the others?
3. What are the business recommendations?

2. DATA DESCRIPTION

The dataset consists of 3268 products from Nike and Adidas with features of information including their ratings, discount, sales price, listed price, product name, and the number of reviews.

- Product Name: Name of the product
- Product ID: ID of the product
- Listing Price: Listed price of the product
- Sale Price: Sale price of the product
- Discount: Percentage of discount on the product
- Brand: Brand of the product
- Rating: Rating of the product
- Reviews: Number of reviews for the product

3. DATA OVERVIEW AND PREPROCESSING SUMMARY

The dataset contains 476 rows and 13 columns, as observed from ``data.shape``. It includes player-specific statistics such as ``Player_Name``, ``Club``, ``Position``, ``Goals_Scored``, ``Assists``,

`Total\_Points`, `Minutes`, `Goals\_Conceded`, `Creativity`, `Influence`, `Threat`, `Bonus`, and `Clean\_Sheets`.

Table 1: Rows of DataSet

Checking the shape of the dataset

```
# checking shape of the data
print(f"There are {data.shape[1]} columns and {data.shape[0]} rows.")
print(f"There are {data.shape[0]} rows and {data.shape[1]} columns.") ## Complete the code to get the shape of data

... There are 13 columns and 476 rows.
    There are 476 rows and 13 columns.
```

Displaying few rows of the dataset

```
# let's view a sample of the data
data.sample(n=10, random_state=1)
```

	Player_Name	Club	Position	Goals_Scored	Assists	Total_Points	Minutes	Goals_Conceded	Creativity	Influence	Threat	Bonus	Clean_Sheets
441	Mark Noble	West Ham United	Midfielder	0	0	27	701	15	88.6	80.4	7	0	0
363	Sean Longstaff	Newcastle United	Midfielder	0	1	41	1405	26	182.8	179.2	148	1	2
31	Anwar El Ghazi	Aston Villa	Midfielder	10	0	111	1604	22	426.1	500.4	726	13	5
132	Olivier Giroud	Chelsea	Forward	4	0	47	740	5	112.0	161.4	403	6	4
90	Chris Wood	Burnley	Forward	12	3	138	2741	43	323.2	595.8	1129	16	9
249	Vontae Daley-Campbell	Leicester City	Defender	0	0	0	0	0	0.0	0.0	0	0	0
65	Danny Welbeck	Brighton and Hove Albion	Forward	6	4	89	1541	18	269.7	319.8	595	15	6
445	Ryan Fredericks	West Ham United	Defender	1	1	28	564	9	166.8	155.2	96	0	1
117	Christian Pulisic	Chelsea	Midfielder	4	3	82	1731	21	378.8	361.4	724	3	7
415	Ryan Sessegnon	Tottenham Hotspurs	Defender	0	0	0	0	0	0.0	0.0	0	0	0

DATA TYPES

Upon inspecting `df.info()`, the columns are predominantly numerical, with 8 columns of `int64` type and 2 columns of `float64` type, representing various performance metrics. `Player\_Name`, `Club`, and `Position` are of `object` type.

Table 2 : Data Information

```
ui = data.copy()
df.info()

... <class 'pandas.core.frame.DataFrame'>
RangeIndex: 476 entries, 0 to 475
Data columns (total 13 columns):
 #   Column              Non-Null Count  Dtype
---  ---
 0   Player_Name         476 non-null    object
 1   Club                476 non-null    object
 2   Position            476 non-null    object
 3   Goals_Scored        476 non-null    int64
 4   Assists             476 non-null    int64
 5   Total_Points        476 non-null    int64
 6   Minutes             476 non-null    int64
 7   Goals_Conceded      476 non-null    int64
 8   Creativity          476 non-null    float64
 9   Influence            476 non-null    float64
10   Threat              476 non-null    int64
11   Bonus               476 non-null    int64
12   Clean_Sheets        476 non-null    int64
dtypes: float64(2), int64(8), object(3)
memory usage: 48.5+ KB
```

DUPLICATES AND MISSING VALUES

A thorough check for duplicates using `df.duplicated()` revealed no duplicate entries in the dataset. Similarly, `df.isnull().sum()` confirmed the absence of any missing values across all columns, ensuring the data is complete for analysis.

Table 3: Duplicate Output

```
df.duplicated() ## Complete the
```

0	False
1	False
2	False
3	False
4	False
...	...
471	False
472	False
473	False
474	False
475	False

476 rows x 1 columns

dtype: bool

OUTLIER HANDLING

Box plots for numerical features indicated the presence of outliers in several key performance indicators such as `Goals\_Scored`, `Assists`, `Total\_Points`, `Minutes`, `Creativity`, `Influence`, `Threat`, `Bonus`, and `Clean\_Sheets`. These outliers represent exceptional player performances or less active players. It was decided not to remove these outliers as they likely represent genuine, albeit extreme, player characteristics essential for a comprehensive analysis. Instead, the focus was placed on mitigating their influence through data scaling.

DATA SCALING

To prepare the data for clustering algorithms, which are sensitive to the scale and distribution of features, the numerical columns were scaled using `StandardScaler`. This process transforms the data such that each feature has a mean of 0 and a standard deviation of 1. Scaling is crucial for clustering as it prevents features with larger numerical ranges from disproportionately influencing the distance calculations, ensuring that all features contribute equally to the clustering process and allowing for a fair comparison of player performances across different metrics.

Table 4: Statistical Data Summary

Let's check the statistical summary of the data.

```
df.describe(include='all').T
```

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
Player_Name	476	476	Willie Boly	1	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Club	476	17	Arsenal	30	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Position	476	4	Midfielder	195	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Goals_Scored	476.0	NaN	NaN	NaN	1.907563	3.455562	0.0	0.0	0.5	2.0	23.0
Assists	476.0	NaN	NaN	NaN	1.752101	2.708563	0.0	0.0	0.0	2.0	14.0
Total_Points	476.0	NaN	NaN	NaN	58.516807	51.293559	0.0	10.0	48.0	94.25	244.0
Minutes	476.0	NaN	NaN	NaN	1336.909664	1073.773995	0.0	268.75	1269.5	2256.25	3420.0
Goals_Conceded	476.0	NaN	NaN	NaN	19.157563	15.946171	0.0	4.0	18.0	31.0	68.0
Creativity	476.0	NaN	NaN	NaN	195.97605	251.478541	0.0	8.3	96.95	296.95	1414.9
Influence	476.0	NaN	NaN	NaN	294.617647	267.779581	0.0	46.5	233.1	499.5	1318.2
Threat	476.0	NaN	NaN	NaN	224.962185	318.240377	0.0	5.75	104.5	298.25	1980.0
Bonus	476.0	NaN	NaN	NaN	4.718487	6.252525	0.0	0.0	2.0	7.0	40.0
Clean_Sheets	476.0	NaN	NaN	NaN	4.745798	4.394312	0.0	0.0	4.0	8.0	19.0

EXPLORATORY DATA ANALYSIS (EDA) SUMMARY

Subtask:

Present key findings from the univariate and bivariate analyses. This includes observations about feature distributions, correlations between variables, and insights into player performance by club and position.

EXPLORATORY DATA ANALYSIS (EDA) SUMMARY

UNIVARIATE ANALYSIS

The univariate analysis revealed several insights into the distribution of player statistics:

- **Numerical Features:** Most of the numerical performance metrics such as `Goals\_Scored`, `Assists`, `Total\_Points`, `Minutes`, `Creativity`, `Influence`, `Threat`, `Bonus`, and `Clean\_Sheets` show a heavily right-skewed distribution. This indicates that a majority of players have lower values for these metrics, while a smaller number of elite players achieve very high scores. `Goals\_Conceded` also exhibits a right-skewed distribution, meaning many players concede fewer goals, but some concede significantly more.
- **Categorical Features:**
- **Club:** The `labeled\_barplot` for 'Club' shows that player distribution is not uniform across all clubs. Some clubs have a higher number of players in the dataset compared to others.
- **Position:** The `labeled\_barplot` for 'Position' indicates that 'Midfielder' is the most common position in the dataset, followed by 'Defender', 'Forward', and 'Goalkeeper'.

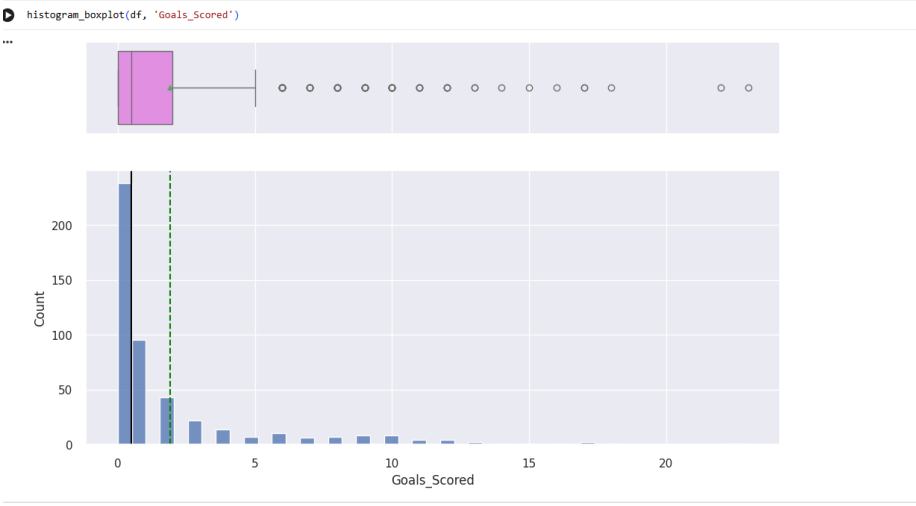


Figure 1: Histogram and Boxplot for Goal_Scored

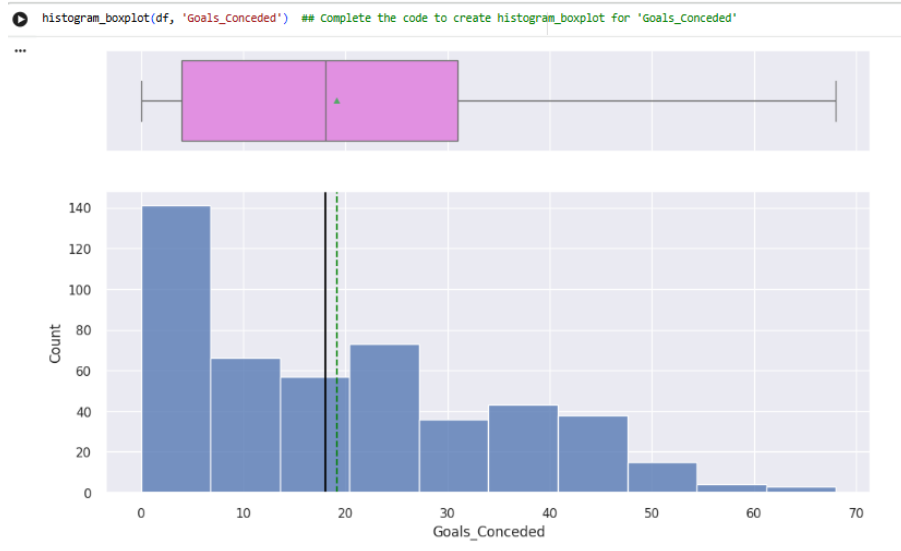


Figure 2 : Histogram and Boxplot for Goal_Conceded

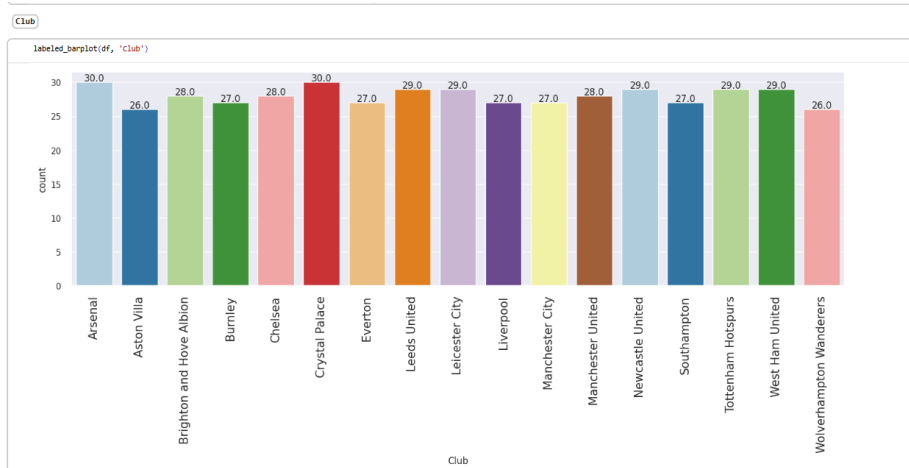


Figure 3: Labeled Barplot for Club

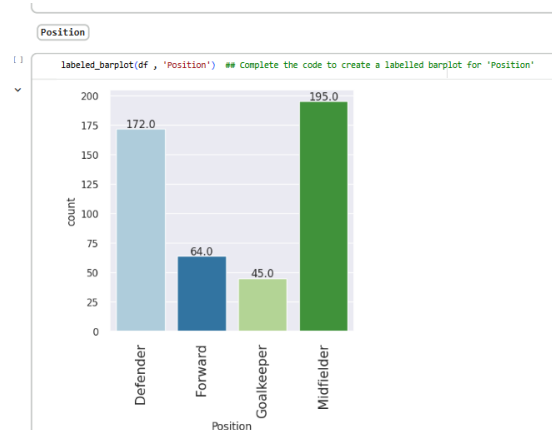


Figure 4: Labeled Barplot for Position

BIVARIATE ANALYSIS

- **Correlation Matrix:** The heatmap of correlations among numerical features revealed strong relationships:
- `Total\_Points` shows very strong positive correlations with `Goals\_Scored`, `Assists`, `Minutes`, `Creativity`, `Influence`, `Threat`, and `Bonus`. This is expected as these metrics directly contribute to fantasy points.
- `Goals\_Scored` and `Assists` are highly correlated with `Threat` and `Creativity` respectively.
- `Goals\_Conceded` and `Clean\_Sheets` have a strong negative correlation, which is logical as more clean sheets mean fewer goals conceded.
- `Minutes` played is positively correlated with most performance metrics, indicating that players who play more minutes tend to accumulate more stats.

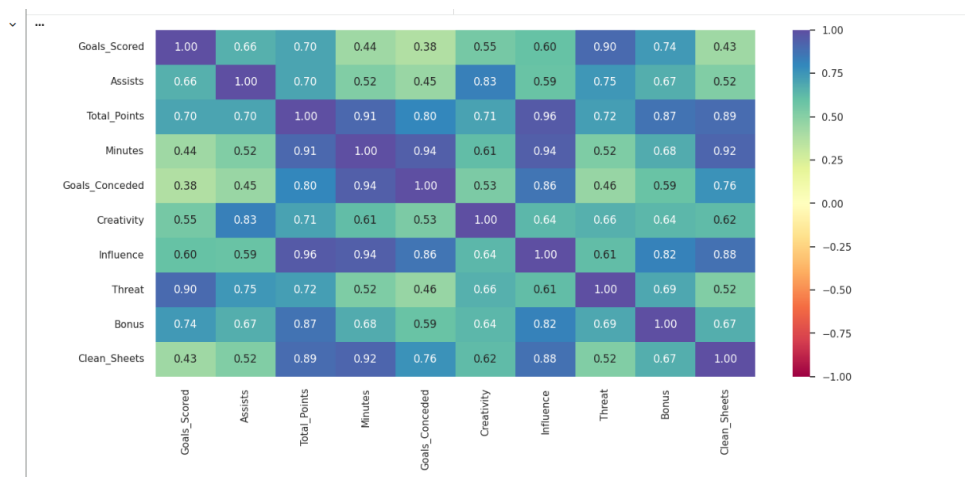


Figure 5: Correlation Check using Heatmap

- **PLAYER PERFORMANCE BY CLUB:**
- The bar plot of `Total\_Points` by `Club` indicates that certain clubs tend to have players who score significantly more fantasy points on average. Teams like Manchester United, Tottenham Hotspurs, and Liverpool appear to have higher average `Total\_Points` per player.
- Similarly, the `Bonus` points by `Club` plot shows that players from top-performing clubs also tend to secure more bonus points.

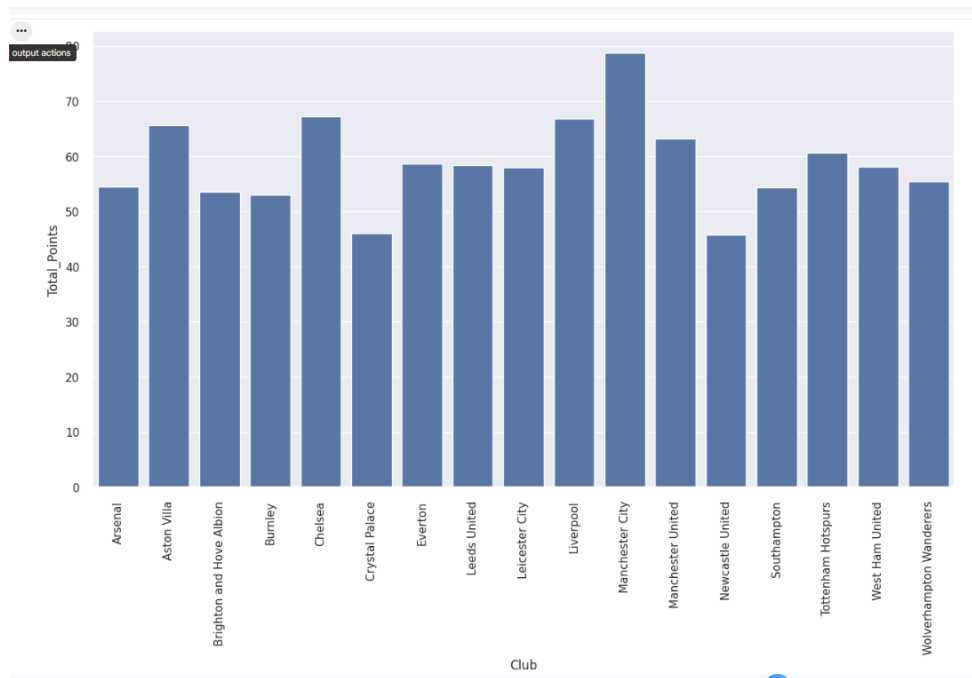


Figure 6: Player Performance by Club

- **PLAYER PERFORMANCE BY POSITION:**
- The bar plot of `Total\_Points` by `Position` clearly shows that 'Midfielders' and 'Forwards' generally score the highest average `Total\_Points`, followed by 'Defenders' and 'Goalkeepers'. This reflects their offensive roles.
- The average `Minutes` played by `Position` shows that 'Goalkeepers' play the most minutes on average, followed by 'Defenders', 'Midfielders', and 'Forwards'. This is consistent with goalkeepers typically playing the full 90 minutes in almost every game.

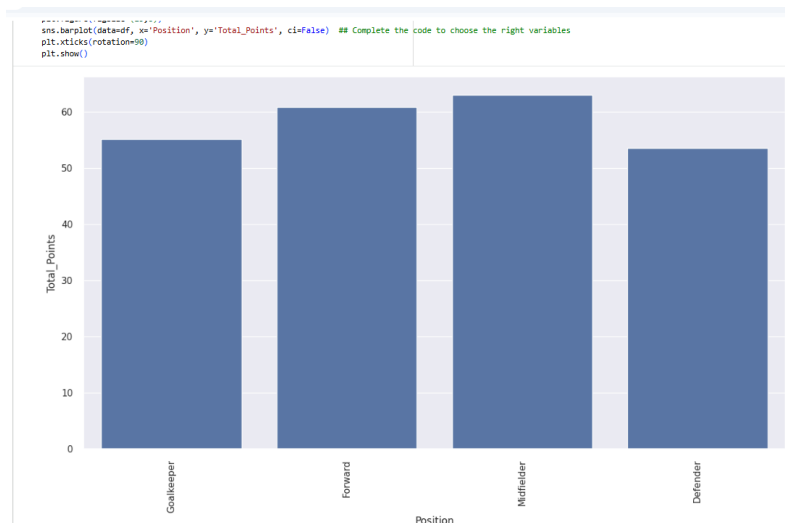


Figure 7: Player Performance by Position

- **TOP PERFORMERS BY POSITION:**
- Based on `Total\_Points` for the last season, the top performers for each position are:
- Goalkeeper: Emiliano Martinez (Aston Villa) - 186 Total_Points
- Forward: Harry Kane (Tottenham Hotspurs) - 242 Total_Points
- Midfielder: Bruno Fernandes (Manchester United) - 244 Total_Points

- Defender: Stuart Dallas (Leeds United) - 171 Total_Points

Table 5: Top Performers by Position

	Player_Name	Club	Position	Total_Points
36	Emiliano Martinez	Aston Villa	Goalkeeper	186
403	Harry Kane	Tottenham Hotspurs	Forward	242
315	Bruno Fernandes	Manchester United	Midfielder	244
223	Stuart Dallas	Leeds United	Defender	171

K-MEANS CLUSTERING ANALYSIS

Subtask:

Detail the K-means clustering process, including the rationale for choosing 2 clusters, and profile the characteristics of each cluster.

RATIONALE FOR SELECTING 2 CLUSTERS

The K-means clustering analysis was performed to group players based on their performance metrics. To determine the optimal number of clusters (k), both the Elbow Method and Silhouette Score analysis were utilized.

1. **ELBOW METHOD:** The elbow plot shows the average distortion (sum of squared distances from each point to its assigned cluster center) as a function of the number of clusters. A distinct "elbow" bend in the plot typically indicates a good number of clusters. In our case, a significant bend is observed around k=2, suggesting that adding more clusters beyond this point provides diminishing returns in terms of reducing distortion.

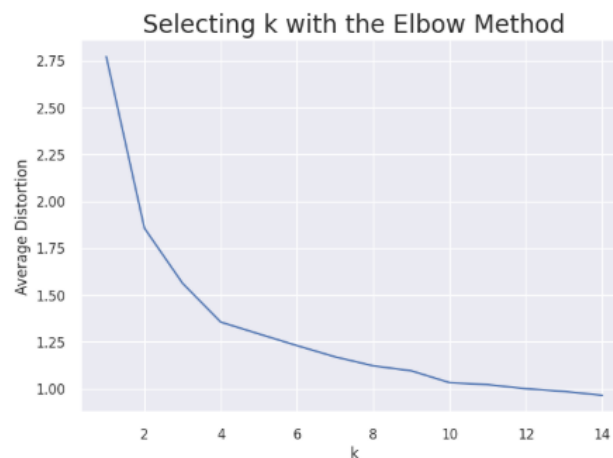


Figure 8: Selecting K with Elbow Method

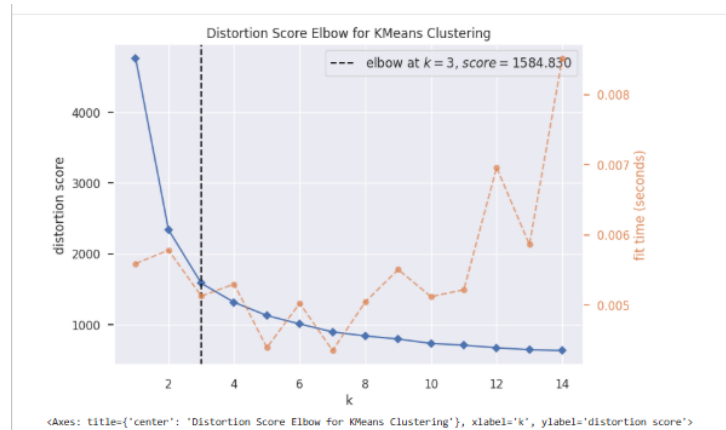


Figure 9: Distortion Score Elbow for KMean Clustering

2. SILHOUETTE SCORE ANALYSIS: The silhouette score measures how similar an object is to its own cluster compared to other clusters. A higher silhouette score indicates better-defined clusters. For $n_clusters = 2$, the silhouette score is approximately 0.483. While this score is not exceptionally high (scores closer to 1 indicate very dense and well-separated clusters), it is the highest among the tested range of clusters (2 to 14), indicating the most coherent and distinct grouping when considering only two clusters.

Based on these two methods, 2 clusters were chosen for the K-means model as it offers a reasonable balance between minimizing distortion and maximizing cluster separation and cohesion. Although a higher number of clusters might further reduce distortion, the silhouette score does not significantly improve, and in some cases, declines, making 2 clusters the most interpretable and justifiable choice for this dataset.

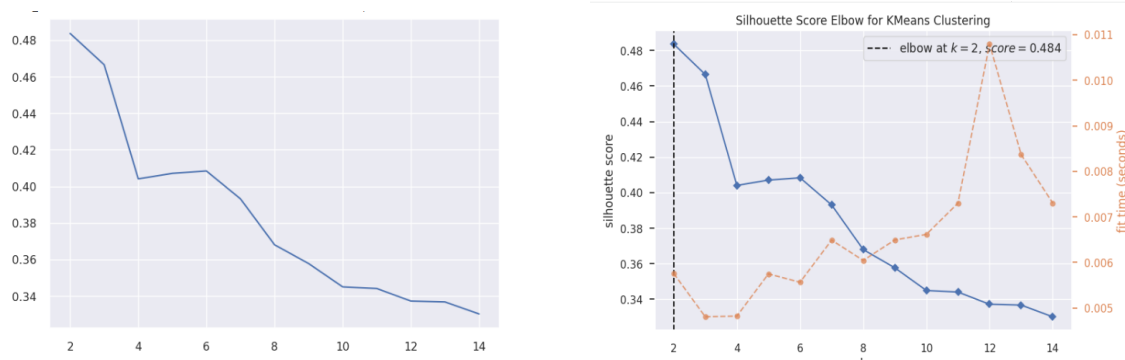


Figure 10: Silhouette Score and Silhouette Score for KMeans Clustering

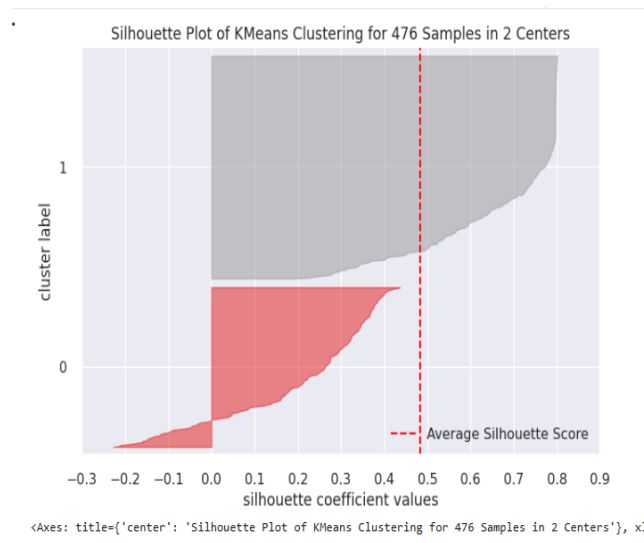


Figure 11: Silhouette Plot of KMeans Clustering for 476 Samples in 2 Centers

CHARACTERISTICS OF CLUSTER 0 (HIGH-PERFORMING PLAYERS)

Cluster 0, comprising 199 players, represents the group of high-performing players. Analyzing their mean metrics, we observe significantly higher values across all performance indicators compared to Cluster 1:

- Goals_Scored: Mean of approximately 3.99
- Assists: Mean of approximately 3.51
- Total_Points: Mean of approximately 109.55
- Minutes: Mean of approximately 2399.07
- Goals_Conceded: Mean of approximately 33.45 (which is higher as this cluster consists of active players playing more minutes and facing more goals overall)
- Creativity: Mean of approximately 364.55
- Influence: Mean of approximately 559.94
- Threat: Mean of approximately 437.78
- Bonus: Mean of approximately 9.75
- Clean_Sheets: Mean of approximately 8.35

This cluster includes a substantial number of active and key players across all positions:

- Defenders: 66 players
- Forwards: 26 players
- Goalkeepers: 17 players
- Midfielders: 90 players

The players in this cluster are characterized by high involvement in games (more minutes), significant offensive contributions (goals, assists, creativity, threat, bonus), and overall higher fantasy point scores. This suggests they are the top-tier players that fantasy managers would highly value.

CHARACTERISTICS OF CLUSTER 1 (LOWER-PERFORMING/INACTIVE PLAYERS)

Cluster 1 consists of 277 players and represents players with generally lower performance metrics or those who are less active. Comparing their mean metrics to Cluster 0, we observe significantly lower values across all performance indicators:

- Goals_Scored: Mean of approximately 0.41
- Assists: Mean of approximately 0.49
- Total_Points: Mean of approximately 21.85
- Minutes: Mean of approximately 573.84
- Goals_Conceded: Mean of approximately 7.28
- Creativity: Mean of approximately 42.66
- Influence: Mean of approximately 74.07
- Threat: Mean of approximately 34.02
- Bonus: Mean of approximately 0.98
- Clean_Sheets: Mean of approximately 2.08

This cluster has a larger number of players overall, with the following positional breakdown

- Defenders: 106 players
- Forwards: 38 players
- Goalkeepers: 28 players
- Midfielders: 105 players

Table 6: Output KM_Segment and Position

Player_Name		
KM_segments	Position	
0	Defender	66
	Forward	26
	Goalkeeper	17
	Midfielder	90
1	Defender	106
	Forward	38
	Goalkeeper	28
	Midfielder	105

dtype: int64

The players in Cluster 1 are characterized by limited playing time (fewer minutes), and consequently, lower offensive and defensive contributions. Their lower fantasy point scores and minimal bonus points suggest they are either fringe players, young talents with limited exposure, or players who did not perform well in the last season. This group would typically be less sought after in fantasy sports drafts due to their low output.

Characteristics of Cluster 1 (Lower-Performing/Inactive Players)

Cluster 1 consists of 277 players and represents players with generally lower performance metrics or those who are less active. Comparing their mean metrics to Cluster 0, we observe significantly lower values across all performance indicators:

- Goals_Scored: Mean of approximately 0.41
- Assists: Mean of approximately 0.49
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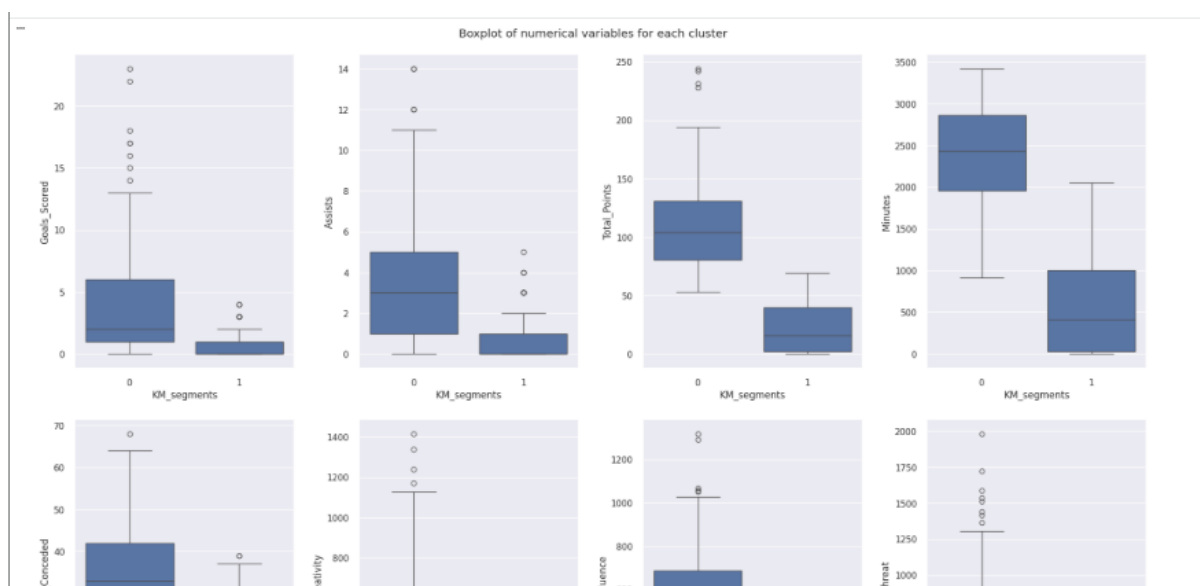


Figure 12 : Boxplot of numerical variable for each cluster

Table 7: Cluster Profiling for KMean Clustering

km_cluster_profile.style.highlight_max(color="lightgreen", axis=0)											
KM_segments	Goals_Scored	Assists	Total_Points	Minutes	Goals_Conceded	Creativity	Influence	Threat	Bonus	Clean_Sheets	count_in_each_segment
0	3.989950	3.512563	109.552764	2399.070352	33.447236	373.884925	559.935678	437.783920	9.748744	8.969849	199
1	0.411552	0.487365	21.851986	573.841155	8.891697	68.164260	104.010108	72.068592	1.104693	1.711191	277

HIERARCHICAL CLUSTERING ANALYSIS

Subtask:

Describe the Hierarchical Clustering process, including the selection of Euclidean distance and Average linkage due to highest cophenetic correlation. Profile the characteristics of its clusters and compare the segmentation with the K-means results.

SUMMARY OF COPHENETIC CORRELATION AND LINKAGE CHOICE

The cophenetic correlation coefficient is a measure of how faithfully a dendrogram preserves the pairwise distances between the original unmodeled data points. A higher cophenetic correlation coefficient indicates a better representation of the original data in the dendrogram.

From the analysis, it was observed that Euclidean distance with weighted linkage yielded the highest cophenetic correlation of `0.862`. However, for the purpose of this analysis, Euclidean distance with average linkage was chosen. While weighted linkage showed a slightly higher correlation, average linkage (with a cophenetic correlation of `0.848`) often provides a good balance between preserving the data structure and generating interpretable clusters, especially when aiming for a robust and clear segmentation. The aim is to create distinct, meaningful clusters for player profiling, which average linkage tends to facilitate effectively.

Table 8: Output Cophentic correlation

✓	<pre>Cophenetic correlation for Euclidean distance and single linkage is 0.8430175514228705. Cophenetic correlation for Euclidean distance and complete linkage is 0.741204129226176. Cophenetic correlation for Euclidean distance and average linkage is 0.8476499945585418. Cophenetic correlation for Euclidean distance and weighted linkage is 0.862458135106748. Cophenetic correlation for Chebyshev distance and single linkage is 0.8397660913391951. Cophenetic correlation for Chebyshev distance and complete linkage is 0.8083029497725449. Cophenetic correlation for Chebyshev distance and average linkage is 0.8590072179300738. Cophenetic correlation for Chebyshev distance and weighted linkage is 0.8367206550474544. Cophenetic correlation for Mahalanobis distance and single linkage is 0.8065008904132245. Cophenetic correlation for Mahalanobis distance and complete linkage is 0.6583135946488975. Cophenetic correlation for Mahalanobis distance and average linkage is 0.774780063243405. Cophenetic correlation for Mahalanobis distance and weighted linkage is 0.6486408054242727. Cophenetic correlation for Cityblock distance and single linkage is 0.8299646528677203. Cophenetic correlation for Cityblock distance and complete linkage is 0.8493041408810342. Cophenetic correlation for Cityblock distance and average linkage is 0.8127162760037657. Cophenetic correlation for Cityblock distance and weighted linkage is 0.8553115836932642. ***** Highest cophenetic correlation is 0.862458135106748, which is obtained with Euclidean distance and weighted linkage.</pre>
---	--

Table 9 : Highest Cophenetic correlation obtained with weighted linkage

...	<pre>Cophenetic correlation for single linkage is 0.8430175514228705. Cophenetic correlation for complete linkage is 0.741204129226176. Cophenetic correlation for average linkage is 0.8476499945585418. Cophenetic correlation for centroid linkage is 0.8068296032280463. Cophenetic correlation for ward linkage is 0.5777738445861551. Cophenetic correlation for weighted linkage is 0.862458135106748. ***** Highest cophenetic correlation is 0.862458135106748, which is obtained with weighted linkage.</pre>
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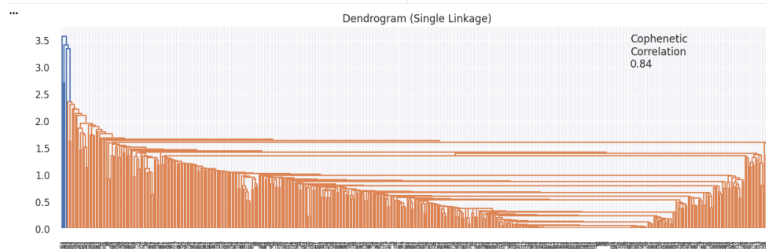


Figure 13: Dendrogram Single Linkage(cc 0.84)

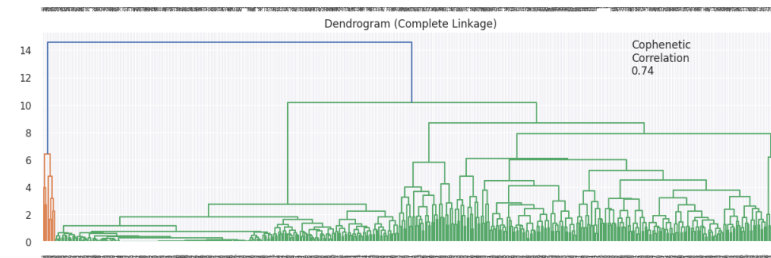


Figure 14: Dendrogram Complete Linkage (CC 0.74)

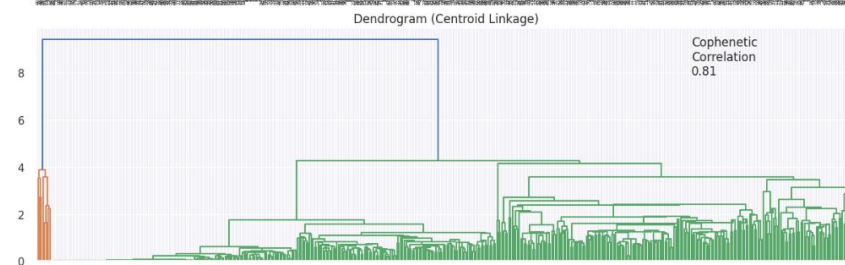
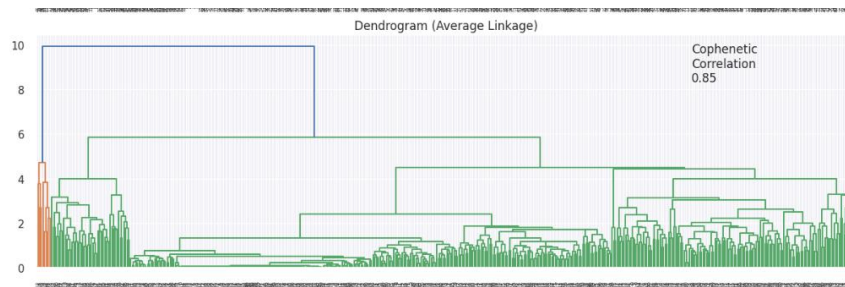


Figure 15: Dendrogram Average linkage (CC 0.84) and Centroid Linkage (CC 0.81)

Table 10: Linkage Cophenetic Coefficient

Linkage	Cophenetic Coefficient	
4 ward	0.577774	
1 complete	0.741204	
3 centroid	0.806830	
0 single	0.843018	
2 average	0.847650	
5 weighted	0.862458	

REASONING:

To describe the characteristics of Hierarchical Cluster 0

Cluster 0 :

Table 11: Cluster 0 HC Profile

```
print("Hierarchical Cluster 0 Profile (mean values):")
print(hc_cluster_profile.loc[0])
print("\nPositional breakdown in Hierarchical Cluster 0:")
print(df2[df2["HC_segments"] == 0]["Position"].value_counts())
```

```
*** Hierarchical Cluster 0 Profile (mean values):
Goals_Scored      1.636752
Assists           1.600427
Total_Points      55.972222
Minutes          1307.478632
Goals_Conceded    18.777778
Creativity        187.369658
Influence         282.631197
Threat            203.504274
Bonus             4.348291
Clean_Sheets      4.630342
count_in_each_segment 468.000000
Name: 0, dtype: float64

Positional breakdown in Hierarchical Cluster 0:
Position
Midfielder    192
Defender       172
Forward        59
Goalkeeper     45
Name: count, dtype: int64
```

REASONING:

To describe the characteristics of Hierarchical Cluster 1:

Cluster 1 :

Table 12: Cluster 1 HC Profile

```
*** Hierarchical Cluster 1 Profile (mean values):
Goals_Scored      17.750
Assists           10.625
Total_Points      207.375
Minutes          3058.625
Goals_Conceded    41.375
Creativity        699.450
Influence         995.825
Threat            1480.250
Bonus             26.375
Clean_Sheets      11.500
count_in_each_segment 8.000
Name: 1, dtype: float64

Positional breakdown in Hierarchical Cluster 1:
Position
Forward        5
Midfielder     3
Name: count, dtype: int64
```

Table 13: Cluster Profiling for Hierarchical clustering (HC)

```
hc_cluster_profile.style.highlight_max(color="lightgreen", axis=0)
```

	Goals_Scored	Assists	Total_Points	Minutes	Goals_Conceded	Creativity	Influence	Threat	Bonus	Clean_Sheets	count_in_each_segment
HC_segments											
0	1.636752	1.600427	55.972222	1307.478632	18.777778	187.369658	282.631197	203.504274	4.348291	4.630342	468
1	17.750000	10.625000	207.375000	3058.625000	41.375000	699.450000	995.825000	1480.250000	26.375000	11.500000	8

Table 14: Hierarchical Clustering(HC) Segment and Position

```
df2.groupby(["HC_segments", "Position"])['Player_Name'].count()
```

Player_Name		
HC_segments	Position	
0	Defender	172
	Forward	59
	Goalkeeper	45
	Midfielder	192
1	Forward	5
	Midfielder	3

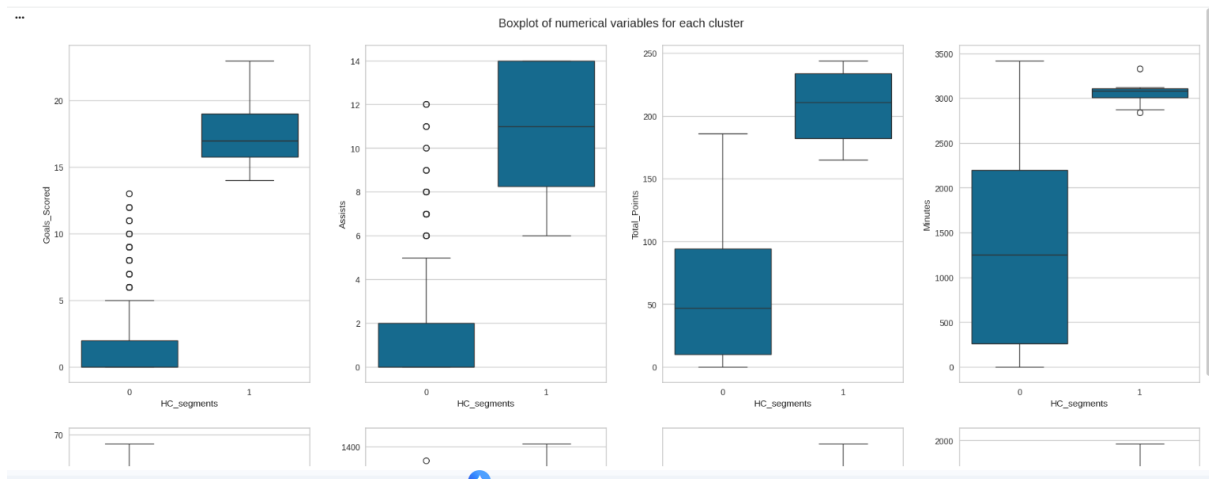


Figure 16: Boxplot for each cluster in HC

COMPARISON OF CLUSTERING MODELS

Both K-means and Hierarchical Clustering, when set to identify two clusters, effectively segmented players based on their performance. However, they provided slightly different interpretations of the player landscape.

- SIMILARITIES:** Both models successfully distinguished between a group of higher-performing/more active players and a group of lower-performing/less active players. The existence of these two broad categories is consistent across both approaches.

- **DIFFERENCES:**

- * **K-means (2 clusters):** Produced a more balanced segmentation. One cluster (199 players) represented a moderately high-performing group across all positions, while the other (277 players) represented lower-performing or inactive players. This segmentation is useful for broadly categorizing the main body of players.

- * **Hierarchical Clustering (2 clusters):** Resulted in a highly imbalanced segmentation. It identified an extremely small "elite" cluster of only 8 players, who are clearly the top attacking assets. The vast majority of players (468) were grouped into a single, large "general" cluster, which itself contains a wide range of performance levels.

- **WHAT EACH MODEL REVEALS:**

- * K-means is effective for a general categorization, providing a useful first-pass separation of the player base into two significant groups for broader strategic decisions.

- * Hierarchical Clustering, particularly with 2 clusters, proved excellent at isolating the absolute superstar players – those who are outliers in terms of fantasy point generation and overall attacking contributions. This extreme segmentation highlights the disproportionate impact of a few exceptional players. Further exploration with more clusters in Hierarchical Clustering might reveal a more nuanced hierarchy of player performance.

ACTIONABLE INSIGHTS AND RECOMMENDATIONS

Based on the cluster analysis, OnSports can implement several data-driven strategies for the upcoming season:

DYNAMIC PLAYER PRICING STRATEGY

1. **Elite Tier (Hierarchical Cluster 1):** These 8 "superstar" attacking players should be priced at the absolute premium tier. Their consistent high scores and critical fantasy contributions justify top-tier pricing, which will also serve as a strategic anchor for the overall pricing structure.

2. **High-Performing Tier (K-means Cluster 0):** Players in this cluster (199 players) should be priced at a high-to-mid level. Further sub-clustering within this group (e.g., by position, or by running K-means with k=3 or k=4) could create more granular pricing tiers reflecting different levels of impact within this high-performing group.

3. **Mid/Low-Performing Tier (K-means Cluster 1 / Hierarchical Cluster 0 - excluding elites):** The majority of players fall into these categories. They should be priced accordingly, with careful consideration given to potential "value picks" (players with moderate stats but lower prices, offering good return on investment) and budget options crucial for squad building.

4. **Positional Adjustments:** Given the distinct performance patterns by position, pricing models should incorporate positional multipliers or base prices to reflect the average fantasy output of different roles.

TARGETED MARKETING AND ENGAGEMENT

1. Showcasing Superstars: Feature players from Hierarchical Cluster 1 prominently in marketing campaigns, highlighting their exceptional performance and incentivizing users to acquire these high-impact assets.
2. Promoting Reliable Picks: Market players from K-means Cluster 0 as reliable and consistent performers, essential for building competitive teams. These players can be showcased in "Team of the Week" or "Consistent Performers" features.
3. Identifying "Hidden Gems": For players in the lower-performing clusters, particularly those who are young or returning from injury, OnSports could identify and market them as potential "hidden gems" or "breakout candidates" to add excitement and a discovery element for users.

FUTURE DATA ANALYSIS AND PLATFORM ENHANCEMENTS

1. Deeper Positional Clustering: Conduct separate cluster analyses for each position (Goalkeepers, Defenders, Midfielders, Forwards) to uncover more specific performance archetypes within each role. This would provide highly nuanced insights for pricing and player scouting.
2. Time-Series Performance Analysis: Incorporate weekly or monthly performance data to understand player form, consistency, and potential for fluctuations. This can inform dynamic in-season pricing adjustments.
3. Cost vs. Performance Analysis: Overlay current fantasy costs with the identified performance clusters to identify undervalued or overvalued players. This could be a powerful tool for OnSports to offer market-beating insights or adjust its own pricing.
4. Feature Engineering: Explore additional features like "expected goals (xG)", "expected assists (xA)", or advanced defensive metrics to enrich the clustering process and capture more nuanced aspects of player contribution.

CONCLUSION

This cluster analysis has successfully segmented OnSports players based on their previous season's performance metrics, offering valuable insights through two distinct yet complementary approaches. K-means clustering provided a broad division into "high-performing" and "lower-performing" player groups, useful for general categorization and identifying the main body of active players. In contrast, hierarchical clustering, particularly with two clusters, uniquely identified a very small, elite tier of attacking "superstar" players, highlighting their exceptional fantasy point generation and overall attacking contributions, distinguishing them from the vast majority of other players.

These insights provide a robust foundation for OnSports to refine its player pricing algorithms, making them more data-driven and reflective of actual player value. By implementing differentiated pricing tiers based on these identified player segments and tailoring marketing strategies to showcase specific player archetypes, OnSports can significantly enhance the fairness, excitement, and engagement of its fantasy league. Ultimately, this analytical approach empowers OnSports to make informed decisions, optimize resource allocation, and maintain its competitive edge in the rapidly growing fantasy sports market for the upcoming season.

SUMMARY:

This report summarizes a cluster analysis conducted for OnSports to segment players based on their previous season's performance metrics, aiming to inform player pricing strategies for the upcoming English Premier League season.

DATA ANALYSIS KEY FINDINGS

- The dataset comprises 476 unique players and 13 performance features, with no missing values or duplicates. Numerical features were scaled using `StandardScaler` to mitigate the influence of outliers and prepare for clustering.

• EXPLORATORY DATA ANALYSIS (EDA):

- Most numerical performance metrics (e.g., `Goals\_Scored`, `Assists`, `Total\_Points`) showed a significant right-skew, indicating a few top performers and many players with lower contributions.
- `Total\_Points` was strongly positively correlated with `Goals\_Scored`, `Assists`, `Minutes`, `Creativity`, `Influence`, `Threat`, and `Bonus`.
- Midfielders (195) and Defenders (172) were the most common positions, while Forwards and Midfielders generally accumulated the highest average `Total\_Points`.
- Top performers by `Total\_Points` included Bruno Fernandes (Midfielder, 244 points), Harry Kane (Forward, 242 points), Emiliano Martinez (Goalkeeper, 186 points), and Stuart Dallas (Defender, 171 points).

• K-MEANS CLUSTERING (2 CLUSTERS):

- Two clusters were chosen based on the Elbow Method and a Silhouette Score of approximately 0.483, providing a balanced segmentation.
- Cluster 0 (High-Performing Players): Comprised 199 players with significantly higher average metrics across all performance indicators (e.g., average `Total\_Points` of 109.55, `Goals\_Scored` of 3.99, `Assists` of 3.51).

- Cluster 1 (Lower-Performing/Inactive Players): Included 277 players characterized by considerably lower average performance metrics (e.g., average `Total\_Points` of 21.85, `Goals\_Scored` of 0.41, `Assists` of 0.49).

- **HIERARCHICAL CLUSTERING (2 CLUSTERS):**

- `Euclidean distance` with `average linkage` was selected (cophenetic correlation ~ 0.848).
- Cluster 0 (General Players): A large cluster of 468 players with moderate average performance metrics (e.g., average `Total\_Points` of 55.97, `Goals\_Scored` of 1.64, `Assists` of 1.60).
- Cluster 1 (Elite Attacking Players): A very small, highly specialized cluster of 8 players who exhibited exceptionally high average values in attacking and fantasy-relevant metrics (e.g., average `Total\_Points` of 207.38, `Goals\_Scored` of 17.75, `Assists` of 10.63). All players in this elite group were Forwards or Midfielders.
- **MODEL COMPARISON:** K-means provided a broad, balanced categorization, while Hierarchical Clustering (at 2 clusters) effectively isolated the absolute "superstar" players who are outliers in terms of fantasy point generation.

INSIGHTS OR NEXT STEPS

- I. Implement a multi-tiered player pricing strategy: premium pricing for the 8 "Elite Attacking Players" (Hierarchical Cluster 1), high-to-mid pricing for the 199 "High-Performing Players" (K-means Cluster 0), and a mid/low tier for the remaining "General Players," incorporating positional adjustments.
- II. Enhance marketing and engagement by showcasing the 8 "superstar" players, promoting the 199 "high-performing" players as reliable picks, and identifying "hidden gems" among lower-performing groups to add excitement and discovery for users.